

STANAG 4285: DK8OK's method in Python, compared with ours

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Nils, thank you for sharing *STANAG 4285: Probing the Ionosphere* and the ESSENTIAL app. Your invitation to port it was too good to resist. Below: what the port does, how it compares with the STANAG sounder we had been running, two small points an independent audit raised about the MATLAB core, and two experiments your work prompted. Questions and corrections are very welcome.

1. Data

- **Receiver:** KiwiSDR (v1.902) on a loop-on-ground antenna, in Arlington MA. Its IQ recordings carry a GPS time stamp on every 512-sample block, so every frame has a GPS time.
- **Signal:** French Navy STANAG 4285 on **8552.6 kHz USB** (FUF, Fort-de-France, about 3200 km), recorded overnight on 2026-10-01 (12-kHz IQ). The interval analysed is **10:40-11:10 UTC**, across local sunrise.
- Some checks also use FUF on 4810 kHz and two other 4285 stations on 5097 and 5760.4 kHz.

2. The port

A function-by-function Python translation of the ESSENTIAL measurement core: channelizer (150-3600 Hz USB select, 1800 Hz to 0 Hz, 9600 S/s), PN80 + RRC (0.20, span 10, 4 sps) reference, acquisition (8-s windows, ± 100 Hz, your quality and contrast gates), adaptive then global straight-line PN clock, fractional-lag rhoIQ at $-2...+4$ ms, and the 64-s Hann time-Doppler / cube analysis. Your constants and thresholds are unchanged. Not ported: the UI, the wideband feed, the empirical-PSF residual fit and the burst correlation.

- A synthetic test (PN80 + random 8PSK, +0.5 ppm clock, second path at +1.2 ms with +0.3 Hz) is recovered correctly.
- 30 minutes of Kiwi IQ run in about 10 s.
- **Independent agreement worth mentioning:** in September we derived the 4285 sync sequence from our own recordings by symbol decisions, without the standard. It matches your PN31 bit string exactly, apart from an overall sign.

The port is not published here. **May I publish it** (with full credit to you, and to ZL2AFP and G3PLX as you credit them)? Until then I can send it privately.

3. Comparison with our own sounder

Our sounder correlates every frame with our measured sync template. Comparing it with your method exposed three faults in ours, all in how it kept time from frame to frame. **We fixed them by adopting your idea:** one straight-line PN clock fitted to the whole run with

your robust refit, and every frame read at that clock at fractional lag. Thank you; it is credited in our code.

After the fix, on the same 30 minutes (Fig. 1):

	your method (port)	ours (fixed)
main-path Doppler, receiver-relative median	-1.0414 Hz	-1.0414 Hz
median-removed Doppler tracks, rms difference	5.3 mHz over 108 windows (64 s Hann, 16-s steps)	
secondary response	+2.29 ms, -10.8 dB	+2.29 ms, -11.3 dB
received frame cadence vs GPS time	-0.038 ppm	-0.064 ppm

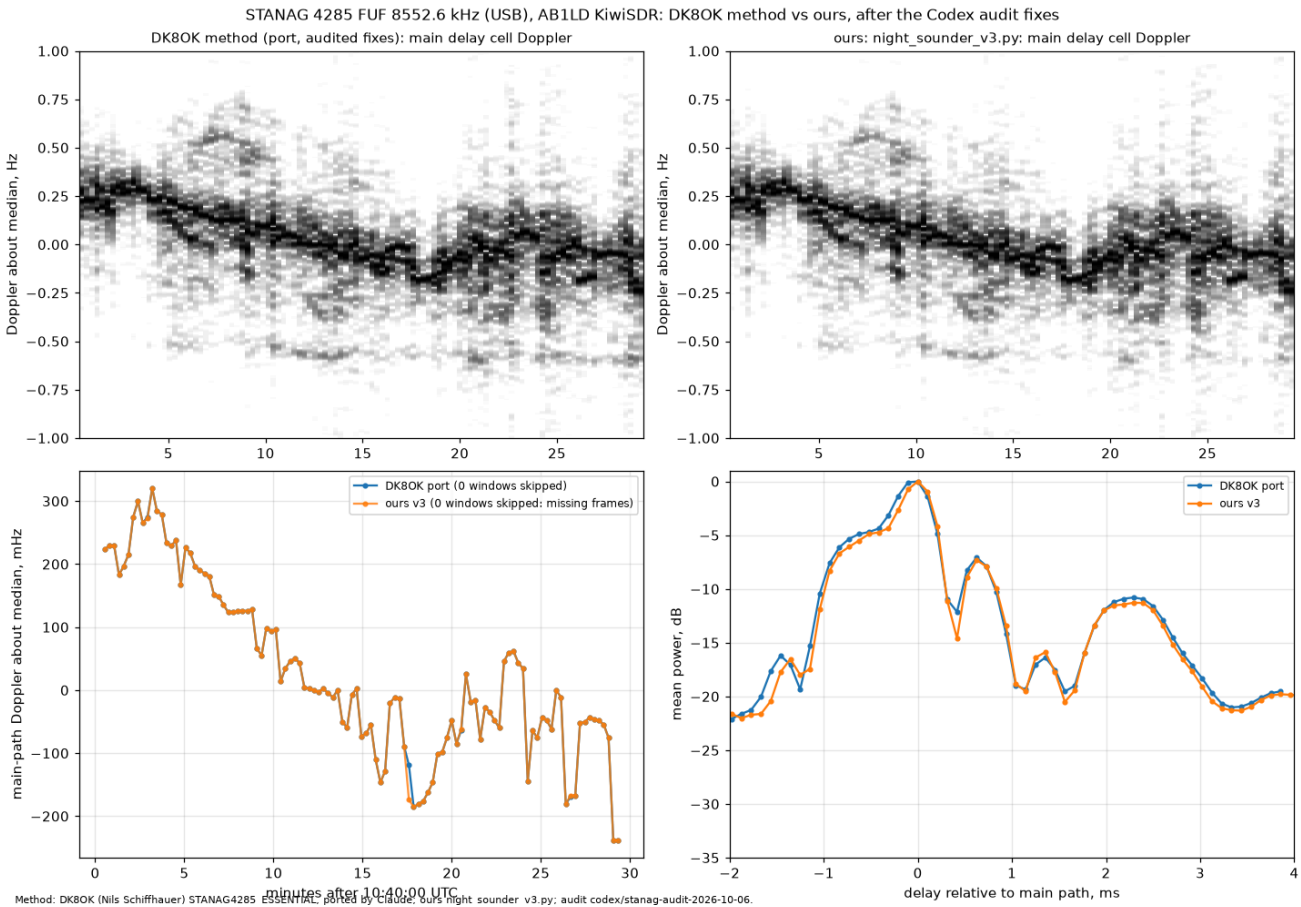


Fig. 1. Main-cell Doppler for each method (top), the two Doppler tracks with their medians removed (lower left), and the mean delay profiles (lower right). FUF 8552.6 kHz, 2026-10-01 10:40-11:10 UTC.

4. Two points from an independent audit of the MATLAB core

We asked Codex (OpenAI) to audit our sounder, the port and the ESSENTIAL measurement core. Most of what it found was in our code. On yours it found the core logic sound: the acquisition gates (peak and contrast), the indexing, the channelizer group-delay compensation and the competing-delay exclusion logic. It raised two points, offered for your consideration. MATLAB was not available to us, so the numbers below come from the port, not your original.

1. **Clock observations are whole samples.** In `v37_residual_core` the peaks used for the clock fit are integer sample positions (`peaks(end+1)=cand(im)`); fractional reading happens only afterwards. On a 240-s synthetic signal with a true +0.5 ppm frame-period offset, the port recovered +0.39 ppm (two paths) and +0.38 ppm (one path). Parabolic interpolation of each peak before the fit recovered +0.49 ppm. This matters only for sub-ppm clock claims.
2. **timingResidualMs** is computed from two straight lines (fitted clock minus the nominal grid). It is the clock's departure from nominal, not the residual of the fit. A name such as `clockDepartureFromNominalMs`, plus the actual peak-minus-fit residuals, might avoid misreading.

A smaller note: linear interpolation of the complex correlation has a slight amplitude ripple between samples (up to about 0.7 dB at half a sample, at the upper band edge). It does not affect Doppler sign or position.

5. Are the bumps real? Sync and probes as independent witnesses

Your *Limits* section warns that sidelobes and overlapping responses can create apparent components. We tried to test that directly (the idea came from Codex). For every frame we estimated the received response by **least squares**, which has no correlation sidelobes, twice:

- from the **80 sync symbols**;
- independently, from the **48 probe symbols** (the three 16-symbol 8PSK blocks, identical in every frame).

The estimates were then used to predict the other set's samples.

- The two estimates agree almost exactly (Fig. 2), including a **shoulder at -0.5 ms**, a **bump at +0.6 ms** and a **component at +2.3 ms**. These are in the received signal, not artefacts of the sync correlation.
- A **bump at +1.3 ms** is about 4 dB weaker in both least-squares estimates than in plain correlation, so it is partly a correlation sidelobe.
- **Held-out prediction:** a channel fitted on the sync predicts the probe samples better when the +1.5...+3.0 ms region is included, in 90 % of frames. A response spanning ± 1 ms predicts about 7 dB better than one spanning ± 0.4 ms.

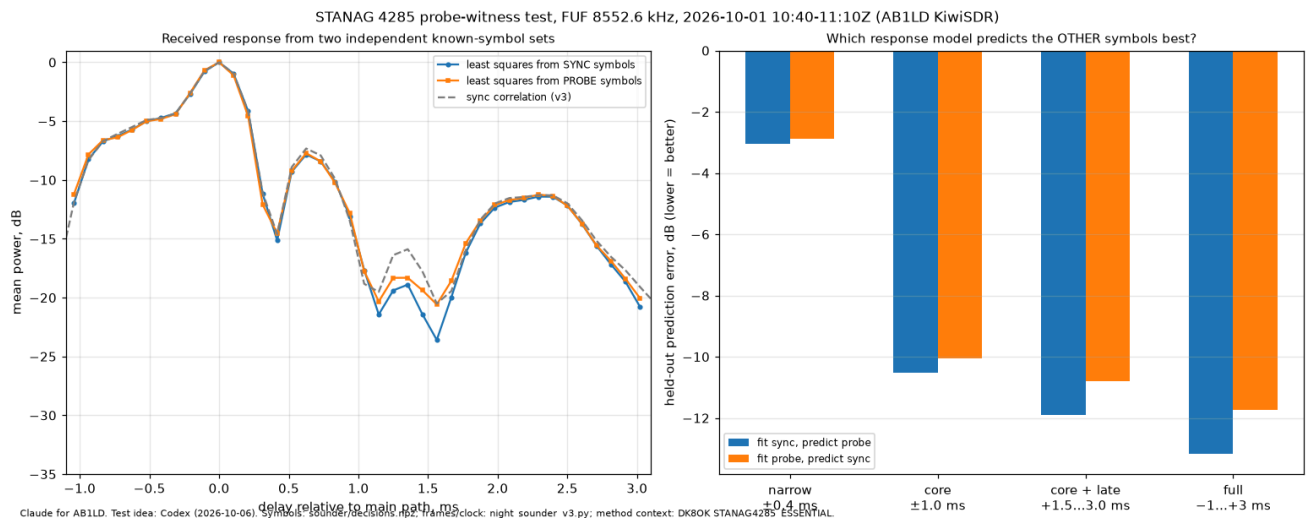


Fig. 2. Left: mean received response by least squares from the sync symbols and, independently, from the probe symbols, with plain sync correlation for comparison. Right: held-out prediction error for four response models.

Which features move with the main path? Each feature's phase relative to the main path (Fig. 3) separates fixed filter or pulse response from independent propagation:

feature	phase relative to main path	relative Doppler	reading
± 0.1 ms (main pulse)	locked	0 Hz	the main path's own response
-0.5 ms	not locked (drifts ≈ -90 cycles in 25 min)	≈ -0.16 Hz	a separate component arriving ≈ 0.5 ms earlier
$+0.6$ ms	mostly locked	mostly 0 Hz	mostly fixed pulse/filter ripple, plus a weak variable part
$+2.3$ ms	not locked ($\approx +120$ cycles)	$\approx +0.4$ Hz, broad	a separate mode; absent at first, strong after $\approx 11:00$ UTC

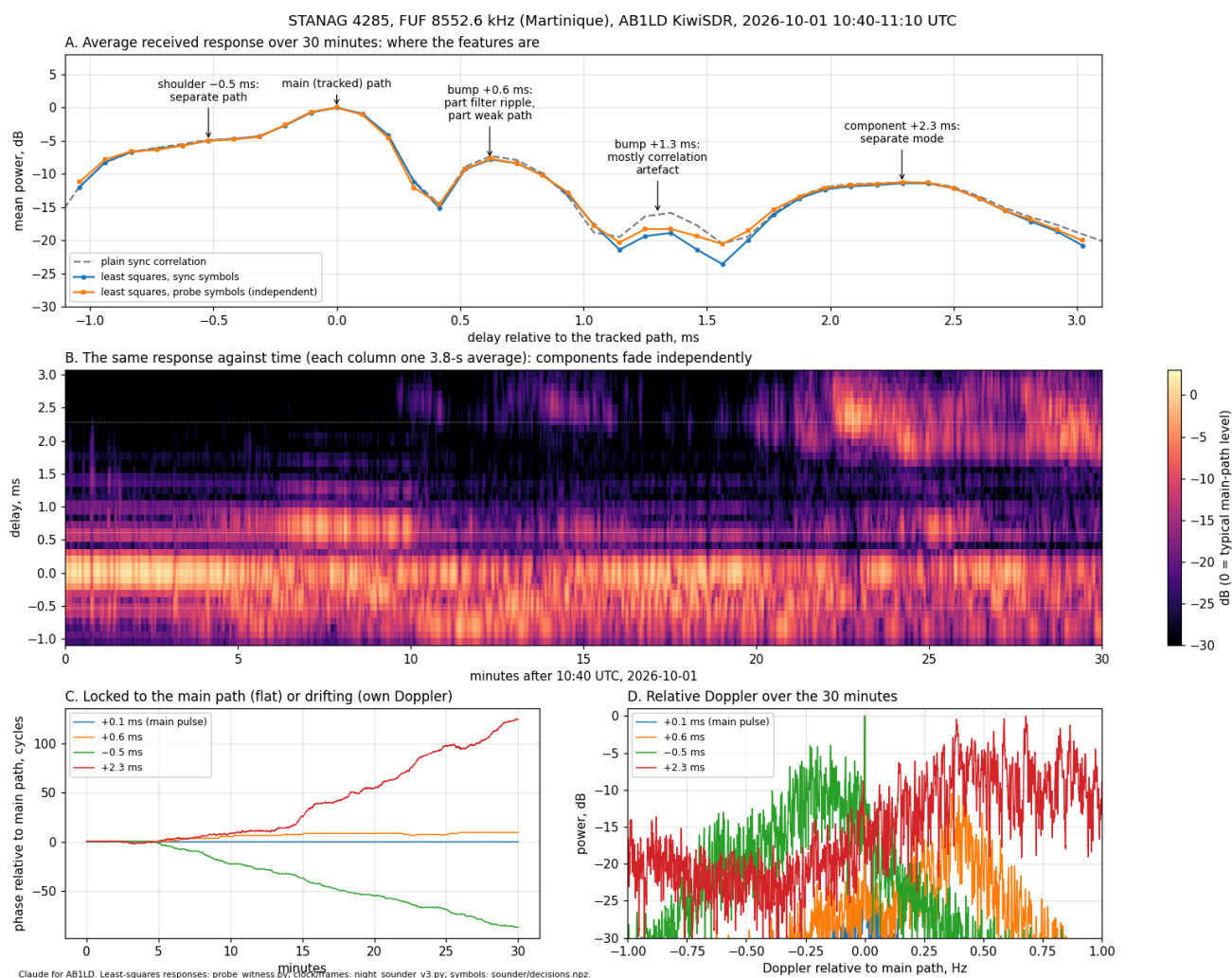


Fig. 3. (A) where the features are; (B) the least-squares response against time; (C) each feature's phase relative to the main path; (D) their Doppler relative to the main path.

Mode identities are **not** established. Plausible candidates for the 3200-km path are one-hop vs two-hop F, low and high rays, or O/X, but we make no claim. We also note that adding the probes as extra *signal* (rather than as a witness) lowered the estimation noise by only 0.7 dB on this strong signal. The probes earn their keep as an independent check.

6. Is FUF's frame timing tied to GPS?

With GPS-stamped samples we can measure where each frame's sync arrives within the 106.667-ms frame, against the UTC second grid (Fig. 4).

- **Two stations are free-running:** 5760.4 kHz drifts about +3.2 ppm, and 5097 kHz about +2.5 ppm with a -9.6 ms jump roughly every 90 minutes.
- **FUF behaves differently**, and its 4810 and 8552.6 kHz transmissions always move together.
- It stayed at 100-104 ms within the frame for ten hours on 10-01.
- It sat near 32-35 ms at two times on 10-01, and at 30.5 ms on 10-06, six days later.
- Then, while we watched, it **jumped +34.5 ms** at 23:54:50 UTC on 10-06, on both frequencies at once.

If FUF's frame timing is tied to GPS/UTC by a fixed rule, we could measure **absolute** group delay rather than relative delay. Do you, or anyone in the group, know whether these transmitters are GPS-disciplined, or what sets their frame epoch?

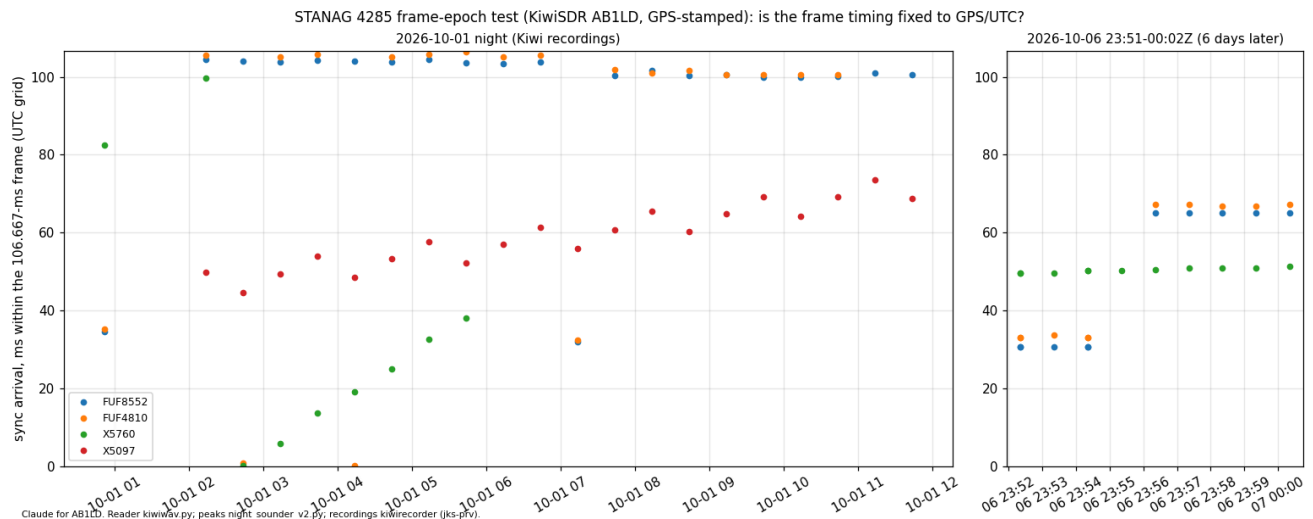


Fig. 4. Sync arrival time within the frame, against the UTC grid, for four STANAG 4285 transmissions. Left: 2026-10-01 night. Right: 2026-10-06, six days later.

7. What we are working on next, and where suggestions would help

These are plans and ideas, not results. Some came from our own work (marked **[AB1LD/Claude]**) and some from the Codex review (marked **[Codex]**). Your comments, and the group's, would be very welcome before we spend effort on them.

Under way

- **Two frequencies, one transmitter** [AB1LD/Claude]. FUF's 4810 and 8552.6 kHz signals share one timing source, so the arrival difference between them (0.6-2.5 ms so far) is a clock-free group-delay difference. With more frequencies at once (we also have a wideband ELAD FDM-S3), this becomes a slice of an oblique ionogram.

Planned

- **Frame-epoch monitoring** [AB1LD/Claude]. A few minutes of FUF every hour (to be scheduled), to catalogue its timing jumps. If the jump sizes are exact multiples of a fixed step and the positions repeat to well under a millisecond, FUF's frame timing is tied to UTC by some rule. Knowing the rule would give **absolute group delay** on this path.
- **Tracking discrete paths instead of reading an averaged profile** [AB1LD/Claude, Codex]. Fit each frame (or a short run of frames) as a few delayed, Doppler-shifted copies of the known symbols plus a diffuse background. A joint delay-Doppler method such as SAGE (Fleury et al., 1999) or CLEAN-style deconvolution separates paths that overlap in delay but differ in Doppler. **Rule:** a component is accepted only if it improves prediction of known symbols that were *not* used to fit it, i.e. the sync/probe test of Section 5 applied every time.
- **Differential observables** [AB1LD/Claude, Codex]. The delay difference and Doppler difference between two components are free of transmitter and receiver clock and frequency errors. Tracking them, with the components' coherence, may be the

cleanest physical product.

- **Phase path versus group path** [AB1LD/Claude]. For each tracked component, compare its Doppler (rate of change of phase path) with the rate of change of its group delay. In a dispersive ionosphere the two differ, and how they differ separates a layer that moves bodily from one whose ionisation is changing.
- **More looks per frame** [Codex]. The sync's 31-symbol repeats are 12.92 ms apart, so their relative phase measures frequency unambiguously over ± 38.7 Hz, versus ± 4.69 Hz from one look per frame. The three probe blocks add three more looks. Together these could settle Doppler aliases and follow fast changes such as the +2.3 ms mode.
- **An event catalogue as the test set** [Codex]. Deep fades, swaps of the strongest mode, abrupt phase changes and transmitter restarts are where a sidelobe and a real path behave differently. We intend to judge methods there, not only on steady, strong reception.

More speculative

- **Using the data symbols** [AB1LD/Claude, Codex]. Decide the 128 data symbols as a modem would (without reading any traffic) and use them as known. Ideally about 5 dB more per frame and a channel estimate across the whole frame, but wrong decisions can create convincing false structure. The probes would stay an independent check.
- **Polarisation** [Codex]. Two coherent receiver channels with different antennas (our SDRplay RSPduo has two tuners on one clock) could show whether the -0.5 ms and $+2.3$ ms components arrive with a different two-antenna response from the main path, a first step towards O/X identification.
- **Several receiving sites** [Codex; AB1LD/Claude]. GPS-timed public KiwiSDRs can record the same transmission at once. We already use them to locate CODAR transmitters by arrival-time differences. For STANAG, a cleaner distant reception could serve as the reference for a weak local one, though the two paths mix in a simple cross-correlation.
- **One sounding core for every signal** [AB1LD/Claude]. Chirp sounders, CODAR and STANAG 4285 are the same measurement: a known waveform, a complex channel response per repetition, Doppler across repetitions. One shared core (GPS-aware reading, clock model with explicit failure, validity masks) would serve all three.
- **Prediction versus observation** [AB1LD/Claude, after your Frequency_Distance app]. Compare ITU-R P.533 predictions with what we measure, starting with an hourly NCDXF/IARU beacon census we have just begun.

Questions for you

1. Is RRC with roll-off 0.20 what you measured for these transmitters, or a standard assumption? A measured pulse shape would let us separate pulse ripple from close paths near ± 0.6 ms.
2. Does the DELUXE empirical one-path PSF ever get estimated from an interval you are confident is single-mode?
3. Which stations and times have you found best for clean, long, single-transmitter runs?
4. Would you, or others in the group, be interested in simultaneous recordings of the same transmission from different sites?

8. Credits

- **Method:** Nils Schiffhauer, DK8OK: *STANAG 4285: Probing the Ionosphere* and DK8OK_STANAG4285_ESSENTIAL (2026). DK8OK credits its origin to Con Wassilieff, ZL2AFP (STANAG 4285 Receiver), and suggestions to Peter Martinez, G3PLX.
- **Audit, the probe-witness idea and several of the ideas in Section 7:** Codex (OpenAI), 2026-10-06.
- **SAGE:** B. H. Fleury et al., IEEE J. Sel. Areas Commun. 17(3), 1999.
- **Receiver software:** KiwiSDR and kiwiclient by John Seamons (ZL4VO/KF6VO) and contributors.
- **Analysis code and drafting:** Claude (Anthropic), working with Mike Burns, AB1LD.

Comments: open an [Issue](#) on this repository, or write in the Dopplergram group.